

# SQL Analysis Report – Sample Sales Data

Lynn Tiffany | AnalystLab Africa — Data Analytics Track | Week 3 Assignment

## 1. Database Setup

**Database:** sales\_analysis **Table:** sales

**Dataset Import:** The dataset was imported into a table named **sales** using the MySQL Workbench Table Data Import Wizard.

**Dataset Overview:** The Sales dataset contains historical sales transactions. Each row represents a product sold within a customer order and includes order details, product information, customer information, sales values, and geographical information.

**Schema Summary:**

- Number of tables: **1**
- Number of Columns: **25**
- Number of Rows: **2823**
- Table name: **sales**
- Relationships: **None (single-table dataset)**
- Primary key: **None defined during import**
- Foreign keys: **None**
- Constraints: **None created automatically during CSV import**

## 2. Core SQL Queries

SQL queries utilizing SELECT, WHERE, ORDER BY, GROUP BY, HAVING, aggregate functions, subqueries, and window functions were developed to answer business-oriented analytical questions. The complete SQL script is provided in the attached sales\_queries.sql file.

| SQL Concept | Purpose                      |
|-------------|------------------------------|
| SELECT      | Retrieve relevant data       |
| WHERE       | Filter records               |
| ORDER BY    | Rank and sort results        |
| GROUP BY    | Aggregate data by categories |
| HAVING      | Filter grouped results       |

|       |                         |
|-------|-------------------------|
| SUM   | Calculate revenue       |
| AVG   | Calculate average sales |
| COUNT | Count transactions      |

### 3. Advanced SQL

**Joins:** Not applicable because the dataset contains a single table.

**Subqueries:** Demonstrated by identifying customers whose total revenue exceeds the average customer revenue.

**Window Functions:** Demonstrated using RANK(), ROW\_NUMBER(), and PARTITION BY to rank customers overall and within each country.

### 4. Business Insights

Which product lines generated the highest revenue?

- **Insight:** Classic Cars generated the highest revenue, indicating they are the company's strongest-performing product category.
- **Recommendation:** Increase inventory and marketing efforts for Classic Cars while evaluating strategies to improve sales of lower-performing product lines.

Who are the Top 10 Customers by Revenue?

- **Insight:** A small group of customers contributes a significant portion of total revenue. Euro shopping channel is the top customer.
- **Recommendation:** Develop loyalty programs and personalized engagement strategies for high-value customers to encourage repeat business.

How did revenue change over time?

- **Insight:** Revenue peaks during specific months, suggesting seasonal demand.
- **Recommendation:** Plan inventory, staffing, and promotional campaigns ahead of peak sales periods.

Customer Purchasing Behavior?

- **Insight:** Some customers place frequent orders, while others make only occasional purchases.
- **Recommendation:** Target infrequent customers with promotions or follow-up campaigns to increase repeat purchases.

## 5. Query Optimization

Query optimization was considered by creating indexes on columns commonly used for filtering and grouping, such as CUSTOMERNAME, ORDERNUMBER, and PRODUCTLINE. Queries were formatted consistently using descriptive aliases, uppercase SQL keywords, and selective column retrieval to improve readability and maintainability.